

International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

Mid-term Examination, Autumn-2023

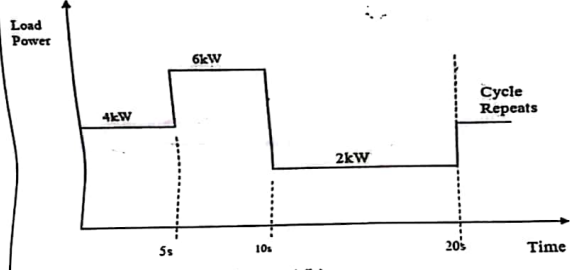
Program: BSc in CSE

Course Title: Electrical Drives and Instrumentation

Course Code: EEE-2421

Time: 1 hour 30 minutes

Full Marks: 30

1(a)	State the faraday's second law. Describe different types of motor duty class with appropriate timing diagram.	CO1	U	3
1(b)	What do you mean by motor sizing? The load torque profile shown in figure 1(b) was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 20%. You do not need to consider starting performance.	CO2	An	3
 Figure: 1(b)				
1(c)	Describe the operation of DC generator with proper net sketch.	CO1	R	4

2(a)	Draw the equivalent circuit of transformer with proper notation.	CO1	U	3
2(b)	When a transformer is connected to a 1000-V, 50-Hz supply the core loss is 1000 W, of which 650 is hysteresis and 350 is eddy current loss. If the applied voltage is raised to 2000 V and the frequency to 100 Hz, find the new core losses.	CO2	An	3
2(c)	In no-load test of single phase transformer, the following test data were obtained: Primary voltage: 220V; Secondary voltage: 110V; Primary Current: 0.5A; Power Input: 30W. Resistance of the primary winding = 0.6 ohm Find the following: i) The turn ratio ii) The magnetizing component of no load current iii) Its working (or loss) component iv) The iron loss.	CO2	An	4

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OR

2(a)	Define transformer. Discuss the working principle of a transformer.	CO1	R	3
2(b)	Why transformer test is needed? Compare the characteristics of an Ideal transformer and a practical transformer.	CO1	U	3
2(c)	A 5kVA 200/1000 V, 50Hz, single-phase transformer gave the following test result: O.C. Test (L.V. Side): 200V, 1.2A, 90W S.C. Test (H.V. Side): 50V, 5A, 110W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO2	An	4

3(a)	Describe the hysteresis losses and eddy current loss for transformer.	CO1	U	2
3(b)	Describe the working principle of linear DC machine as a Motor with appropriate figure and graph.	CO1	R	4
3(c)	A 30kVA, 2400/120-V, 50-Hz transformer has a high voltage winding resistance of 0.1 Ω and a leakage resistance of 0.22 Ω . The low voltage winding resistance is 0.035 Ω and the leakage resistance is 0.012 Ω . Find the equivalent winding resistance, reactance and impedance referred to the i) high voltage side and ii) low voltage side	CO2	An	4

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Mid-term Exam, Spring 2023

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Duration: 30 minutes

Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns.

1	a)	Analyze Linear DC Machine as a Motor.	CO1	E	4
1	b)	Explain hysteresis loss and eddy current loss in a ferromagnetic core?	CO1	U	3
1	c)	List the reasons for which it is important to size a motor? Discuss How to choose the right motor briefly?	CO1	Ap	3
2	a)	Compare the characteristics of an Ideal and a practical transformer. Briefly Discuss the steps to eliminate the losses of a transformer?	CO2	U	3
2	b)	A 2200/200V transformer draws a no-load current of 0.6A and absorbs 400 watts. Find the magnetizing and iron loss currents.	CO2	An	3
2	c)	A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results: O.C. Test (L.V. Side): 200 V, 1.2 A, 90 W S.C. Test (H.V. Side): 50 V, 5A, 110 W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO2	An	4

Or

2	a)	Discuss the working principle of a transformer and draw a vector diagram of an ideal transformer?	CO1	U	3
2	b)	A 50-kVA 440/220 V transformer has $R_1 = 3.45 \Omega$, $R_2 = 0.009 \Omega$. The primary reactances are $X_1 = 5.2 \Omega$ and $X_2 = 0.015 \Omega$. i) Calculate the transformer ii) Calculate the resistance as referred to primary iii) Calculate the resistance as referred to secondary iv) Calculate the reactance as referred to both primary and secondary v) Calculate the reactance as referred to both primary and secondary	CO2	An	4

2	a)	In no-load test of a transformer, the following test data were obtained: Primary Voltage: 220V; Secondary voltage: 110V Power input: 30W; Primary Current: 0.5A Find the following: i) The turns ratio ii) Magnetising component of no-load current iii) Its working component iv) Iron loss.	CO2	An	3
2	b)	Write a short note on the basic principle of a single loop DC generator.	CO1	U	2
3	b)	Briefly Discuss on armature reaction which occurs in electrical machines with necessary diagrams.	CO1	U	4
3	c)	A 250-V shunt motor runs at 1000 r.p.m. at no-load and takes 8A. The total armature and shunt field resistances are respectively 0.2Ω and 250Ω . Calculate the speed when loaded and take 50 A. Assume the flux to be constant.	CO2	An	4

- 3(a) "We cannot think of our modern power system without transformer." U CO2 2
Justify the statement.
- 3(b) Briefly explain the different types of losses in a transformer. Draw a three phase delta-wye connection. R CO2 3
- 3(c) The open circuit test and short circuit test were performed on the primary side of a 20-KVA, 8000/240-V, 50-Hz transformer and the following data were taken: An CO1 5

Open circuit test (on primary)

$$V_{oc} = 6000 \text{ V}$$

$$I_{oc} = 7.74 \text{ A}$$

$$P_{oc} = 300 \text{ W}$$

Short circuit test (on primary)

$$V_{sc} = 367 \text{ V}$$

$$I_{sc} = 2.5 \text{ A}$$

$$P_{sc} = 180 \text{ W}$$

Find the impedances of the approximate equivalent circuit referred to the primary side and sketch the circuit.

OR

- 3(c) A 100-KVA, 2000/440-V transformer has $R_1 = 3 \Omega$, $R_2 = 0.07 \Omega$. An CO1 5
The values of reactances are $X_1 = 5.2 \Omega$, $X_2 = 0.075 \Omega$.
i) Calculate equivalent impedance referred to primary.
ii) Calculate equivalent impedance referred to secondary.

R_1 R_2'

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- 1(a) State Fleming Left Hand Law. Why Motor sizing is important? U CO2 2
- 1(b) Describe the different types of motor duty class with proper timing diagram. R CO2 4
- 1(c) State the four basic principle of magnetic fields applied on electrical machines and justify the statement - 'Magnetic fields are the fundamental mechanism by which energy is converted from one form to another in transformer'. An CO1 4
- 2(a) "A DC machine can work as both the generator and the motor" -Justify the statement. U CO2 2
- 2(b) The load torque profile shown in Fig. 2(b) is measured for a constant speed load. Choose an appropriately sized motor to drive this load. (Assume a service factor of 25%). You do not need to consider starting performance. An CO1 3

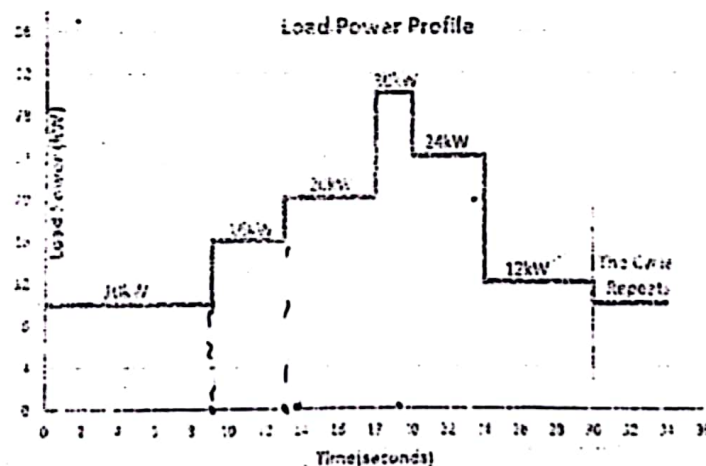


Fig. 2(b)

- 2(c) "A DC generator can be made from an AC generator" Describe the operation of DC generator with neat sketch and also justify the statement. An CO1 5
- OR
- 2(c) Draw the equivalent circuit of a transformer with proper notation. You build a 220V/9V transformer by yourself and want to draw an equivalent circuit of your transformer. How do you get values of all the parameters of the equivalent circuit? Write in detail. An CO1 5